How Accurate Is a Local RAG Pipeline and Can We Even Measure It?

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Abstract

This report documents a small, fully local retrieval-augmented generation (RAG) pipeline built to answer two questions: how accurate is retrieval and generation in such a system, and is there even a reliable way to measure that accuracy? Using Llama 3.1 8B as the generator and TriviaQA's own evidence documents as the retrieval corpus, I measured retrieval accuracy (Recall@k) and answer accuracy (Exact Match and an LLM judge) independently across 90 questions, comparing a retrieval-augmented condition against a no-retrieval baseline. Retrieval-augmented generation outperformed the baseline on every metric, though at only 90 questions the answer-accuracy gap does not clear a conventional significance threshold. Each accuracy metric also turned out to carry its own bias: Exact Match under counts correct paraphrases, and the LLM judge over-credits some answers that are wrong or vague. A manual review of judge decisions found it too lenient in 3 of 18 disagreement cases, suggesting reported judge accuracy is a few points optimistic. The main lesson is that no single metric here tells the whole story.

1 Introduction

Retrieval-augmented generation is usually justified by the claim that grounding a language model's answers in retrieved documents makes them more accurate. This project treats that claim as a question to test directly, on a small scale, using only local and free tools. Two things motivated the design. First, "accuracy" in a RAG system is really two separate questions: did retrieval find the right information, and did the model use it correctly? Conflating them hides which part is failing. Second, what counts as a "correct" passage when ground truth is coarse, and what counts as a "correct" answer when the model paraphrases? This report builds a pipeline simple enough to inspect end to end.

2 Methodology

2.1 Data

Questions and their evidence documents came from TriviaQA (`mandarjoshi/trivia_qa`, config `rc`, validation split), which restricts evidence to documents that contain the answer. 90 questions were sampled with a fixed random seed, each carrying on average about five evidence documents. Each document was split into passages of approximately 175 words with a 20-word overlap, producing 7,853 passages in total. A passage was labeled *correct* for a question if it contained any of the question's accepted answer aliases (case-insensitive, whitespace-normalized).

This labeling is itself an approximation, and not always a mild one. TriviaQA's alias lists sometimes include very short or generic strings (single letters, symbols), which for 6 of the 90 questions made *every* passage in that question's pool count as correct by construction, regardless of content; retrieval accuracy for those 6 questions is trivially perfect and does not reflect retrieval skill. The rule fails in the other direction too: for 2 of the 90 questions, no passage contained any alias at all, so retrieval could not succeed by this measure no matter what it returned. More generally, a passage can be labeled correct by containing the right string without actually explaining the answer, or a genuinely relevant passage can be missed if it phrases the answer differently than the alias list expects.

2.2 Retrieval

Each question and its passages were embedded with `all-MiniLM-L6-v2` (sentence-transformers), and passages were ranked per question by cosine similarity; retrieval was scoped to each question's own evidence passages rather than a global index. Retrieval accuracy was measured as Recall@k: the fraction of questions for which a correct passage appeared in the top k results.

2.3 Generation and scoring

Llama 3.1 8B Instruct (4-bit quantized, served locally via Ollama) generated answers under two conditions. In the *baseline* condition, the model saw only the question and an instruction to answer briefly. In the *RAG* condition, the same instruction was accompanied by the top 5 retrieved passages as context. Both conditions used greedy decoding (temperature 0) for reproducibility.

Each answer was scored two ways. *Exact Match* normalizes the answer (lowercase, strip punctuation, drop articles) and checks it against the normalized answer aliases, a strict, standard metric that penalizes correct paraphrases. *Judge* accuracy uses a second, smaller local model (Phi-3-mini) prompted with the question, the gold answer and its aliases, and the candidate answer, asked to reply yes or no on whether the candidate matches in meaning. Using a separate, smaller model as judge (rather than reusing the generator) avoids a model judging its own output.

To sanity-check the judge, all cases where Exact Match and the judge disagreed were pulled out (23 of 180 answers); 18 were reviewed by hand against my own reading of correctness. 18 was a sample size fixed in advance, not a number chosen because it seemed sufficient, so the remaining 5 disagreements were not reviewed and could in principle include cases the reviewed sample does not represent.

3 Results

3.1 Retrieval accuracy

	Recall@1	Recall@3	Recall@5
Embedding retrieval	0.767	0.867	0.900
Random baseline (simulated)	0.407	0.646	0.740

Table 1: Retrieval accuracy. The random baseline is a Monte Carlo simulation (500 trials) of drawing k passages uniformly at random from each question's own pool and checking whether any is correct.

The chance baseline is the part worth dwelling on. The flat rate of individual passages containing a correct alias is 31.3%, and comparing against that number makes embedding retrieval look roughly three times better than chance at $k = 5$. That comparison is wrong: the task is not "is this one passage correct" but "is any of k passages correct," and drawing 5 passages at random from a pool where nearly a third are labeled correct succeeds most of the time. Simulating the actual k-draw puts chance at 0.407/0.646/0.740 for $k = 1/3/5$. Embedding retrieval still beats chance at every k – by roughly 36, 22, and 16 points respectively – but the margin narrows sharply as k grows, and the headline Recall@5 of 0.900 is much less impressive than it first appears. Together with the 6 degenerate-ground-truth questions noted above, retrieval accuracy on this dataset is real, but neither the baseline nor the ground truth is as clean as it looks.

3.2 Answer accuracy

Condition	Exact Match	Judge	EM/Judge disagreement
Baseline (no retrieval)	65.6%	80.0%	14.4%
RAG (top-5 retrieved)	77.8%	88.9%	11.1%

Table 2: Answer accuracy by condition and metric. RAG improves on the baseline by 12.2 points under Exact Match and 8.9 points under the judge. An exact McNemar test on the paired per-question outcomes gives $p = 0.053$ for Exact Match and $p = 0.115$ for the judge, suggestive but not significant at the conventional 0.05 threshold, given only 90 questions.

The improvement looks slightly larger under Exact Match (+12.2 points) than under the judge (+8.9 points). This is not because retrieval helped Exact Match more, but because the judge already recovers some of the baseline's paraphrased-but-correct answers that Exact Match was missing, which shrinks the apparent baseline deficit.

Every disagreement between Exact Match and the judge (23 of 180 answers, both conditions combined) ran the same direction: the judge credited an answer that Exact Match rejected, never the reverse. This is expected, since Exact Match cannot recognize a correct paraphrase, but it means every judge/EM gap in Table 2 is a judge-leniency gap, not a mix of both.

3.3 Judge reliability spot-check

Of the 18 hand-reviewed disagreement cases, I agreed with the judge's verdict in 15 (83.3%). The three disagreements are shown in Table 3; in every case the judge over-credited an answer that was wrong or too vague, never the reverse.

Question	Gold answer	Candidate	Why the judge was wrong
What does Quitline help people quit	Tobacco or alcohol	"addiction and behavior change/issues"	Too vague to count as the specific answer
Fraction line, written diagonally (baseline run)	Solidus	Diagonal line	Just repeats the question's wording
Activity behind a 'tell' / poker face	Poker	Acting	Wrong activity entirely

Table 3: All three cases, of the 18 reviewed, where I disagreed with the judge.

Extrapolating this 3-in-18 false-positive rate across all 23 disagreements suggests roughly 2% of the 180 total answers may be over-credited by the judge, meaning the Judge accuracy figures in Table 2 are likely a couple of points optimistic in absolute terms. The relative RAG-over-baseline improvement is unaffected, since the same judge scored both conditions. This extrapolation itself rests on only 18 reviewed cases and should be read as a rough order of magnitude, not a precise correction.

4 Conclusion

A local RAG pipeline built from TriviaQA, Llama 3.1 8B, and a smaller judge model shows a positive effect of retrieval on answer accuracy: 65.6% to 77.8% under Exact Match, and 80.0% to 88.9% under an LLM judge. At 90 questions, that effect doesn't clear conventional significance ($p = 0.053$ and $p = 0.115$), so it's a direction worth trusting more than a result worth citing.

The main lesson is that measuring accuracy here takes more than one number, one pass, or one reviewer. Retrieval needs a correct chance baseline, answer accuracy needs both a strict and a lenient metric, and any automated judge needs a manual second look. The case for the second pass is not theoretical: an early draft of this work scored Exact Match against the wrong alias field and compared retrieval against a chance baseline that was far too low. Both were caught only by going back over the numbers a second time, and a larger run I could not have hand-audited would have reported them at face value.

The thing I keep coming back to is what my setup didn't test. TriviaQA's evidence documents are guaranteed to contain the answer, and I scoped retrieval to each question's own passages, so my retriever never had to search a corpus where the answer might not be there, it only had to rank a pool where it was. Real systems don't get that. I'd want to know how much of the reported accuracy of commercial systems survives that difference, and how they handle the case where the right passage simply doesn't exist.

This was a fun first project, and I expect the next one will be less wrong in more interesting ways. If you are ever looking to collaborate, my most recent contact information is on my website: `https://shoimya.github.io/`.